

(1)在/root 这个目录下建立一个名为 vitest 的目录。

图 1

```
[root@localhost ~]# mkdir vitest
[root@localhost ~]# ls
anaconda-ks.cfg  auto_source.sh  initial-setup-ks.cfg  vitest
[root@localhost ~]#
```

图 1 在 root 下创建了一个 vitest 目录

(2)进入 vitest 这个目录当中,将/etc/manpath. config 复制到当前目录。

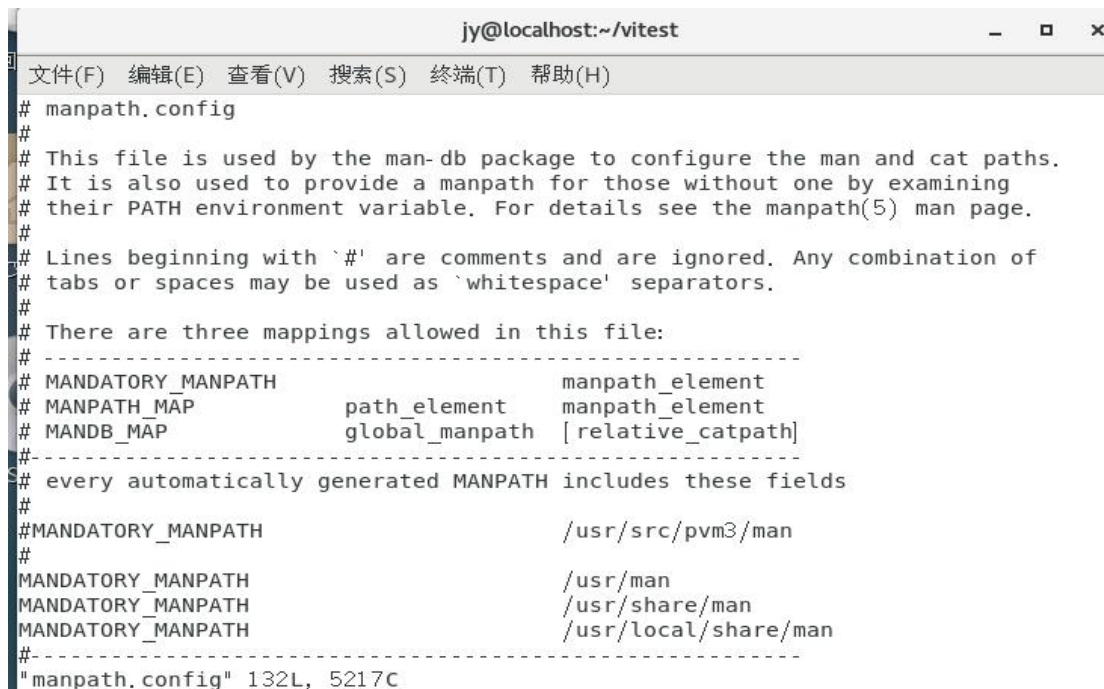
图 2

```
[root@localhost ~]# cd vitest
[root@localhost vitest]# cp /etc/manpath.config .
[root@localhost vitest]# ls
manpath.config
[root@localhost vitest]#
```

图 2 将/etc/manpath.config 复制到创建的 vitest 目录中

(3)使用 vi 打开当前目录下的 manpath. config。

图 3



```
jy@localhost:~/vitest
文件(F) 编辑(E) 查看(V) 搜索(S) 终端(T) 帮助(H)
# manpath.config
#
# This file is used by the man-db package to configure the man and cat paths.
# It is also used to provide a manpath for those without one by examining
# their PATH environment variable. For details see the manpath(5) man page.
#
# Lines beginning with '#' are comments and are ignored. Any combination of
# tabs or spaces may be used as 'whitespace' separators.
#
# There are three mappings allowed in this file:
# -----
# MANDATORY_MANPATH      manpath_element
# MANPATH_MAP             path_element  manpath_element
# MANDB_MAP               global_manpath [relative_catpath]
# -----
# every automatically generated MANPATH includes these fields
#
#MANDATORY_MANPATH      /usr/src/pvm3/man
#
#MANDATORY_MANPATH      /usr/man
#MANDATORY_MANPATH      /usr/share/man
#MANDATORY_MANPATH      /usr/local/share/man
# -----
"manpath.config" 132L, 5217C
```

图 3 使用 vim 打开目录下的 manpath.config 文件

(4)在 vi 中设置行号。

图 4

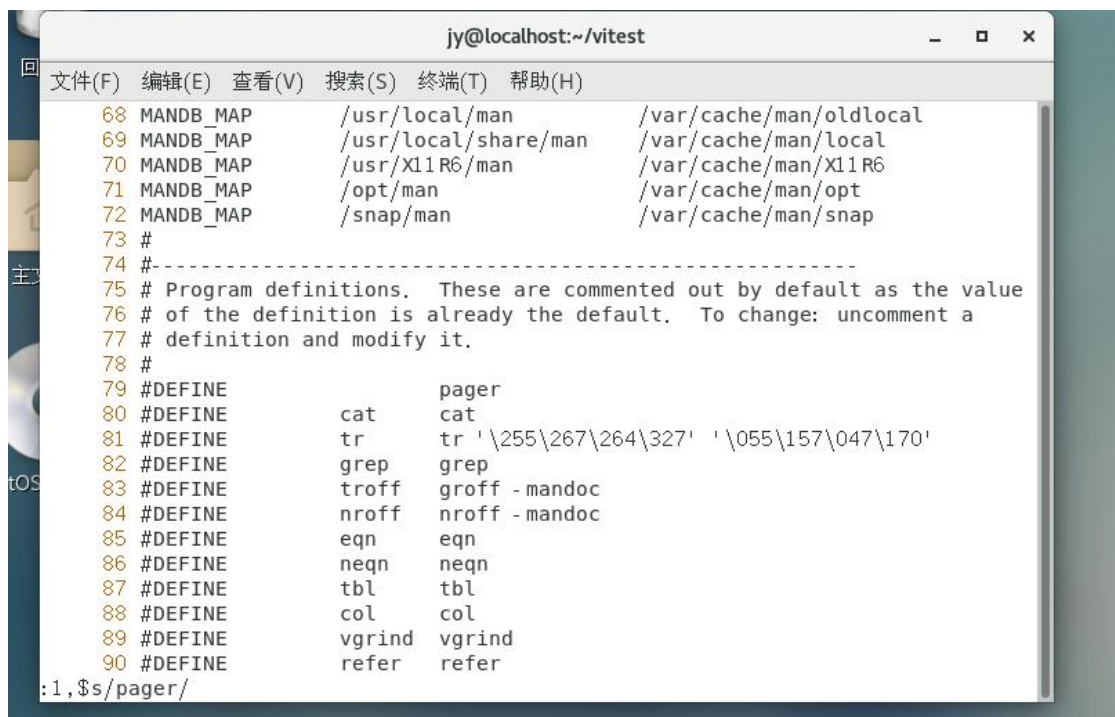


```
1 manpath.config
2 #
3 # This file is used by the man-db package to configure the man and cat p
4 # It is also used to provide a manpath for those without one by examin
5 # their PATH environment variable. For details see the manpath(5) man pa
6 #
7 # Lines beginning with '#' are comments and are ignored. Any combination
8 # of
9 # tabs or spaces may be used as 'whitespace' separators.
10 # There are three mappings allowed in this file:
11 # -----
12 # MANDATORY_MANPATH                manpath_element
13 # MANPATH_MAP                      path_element  manpath_element
14 # MANDB_MAP                        global_manpath [relative_catpath]
15 # -----
16 # every automatically generated MANPATH includes these fields
17 #
18 #MANDATORY_MANPATH                /usr/src/pvm3/man
19 #
: set nu
```

图 4 使用 set nu 命令给这个文件设置了行号

(5) 移动到第一行, 并且向下搜索 pager 这个字符串, 请问它在第几行?

图 5



```
68 MANDB_MAP      /usr/local/man      /var/cache/man/oldlocal
69 MANDB_MAP      /usr/local/share/man  /var/cache/man/local
70 MANDB_MAP      /usr/X11R6/man      /var/cache/man/X11R6
71 MANDB_MAP      /opt/man            /var/cache/man/opt
72 MANDB_MAP      /snap/man           /var/cache/man/snap
73 #
74 # -----
75 # Program definitions. These are commented out by default as the value
76 # of the definition is already the default. To change: uncomment a
77 # definition and modify it.
78 #
79 #DEFINE        pager
80 #DEFINE        cat      cat
81 #DEFINE        tr       tr '\255\267\264\327' '\055\157\047\170'
82 #DEFINE        grep     grep
83 #DEFINE        troff    groff - mandoc
84 #DEFINE        nroff    nroff - mandoc
85 #DEFINE        eqn      eqn
86 #DEFINE        neqn     neqn
87 #DEFINE        tbl      tbl
88 #DEFINE        col      col
89 #DEFINE        vgrind   vgrind
90 #DEFINE        refer    refer
:1,$s/pager/
```

图 5 使用: 1, \$s/pager 命令先移动到第一行再向下搜索了 pager 这个字符串

(6) 接下来, 要将 50~100 行的 man 改为 MAN, 并且一个一个地选择是否需要修改。

图 6

```

jy@localhost:~/vitest
文件(F) 编辑(E) 查看(V) 搜索(S) 终端(T) 帮助(H)
43 # For a manpath element to be treated as a system manpath (as most of th
    ose
44 # above should normally be), it must be mentioned below. Each line may h
    ave
45 # an optional extra string indicating the catpath associated with the
46 # manpath. If no catpath string is used, the catpath will default to the
47 # given manpath.
48 #
49 # You *must* provide all system manpaths, including manpaths for alterna
    te
50 # operating systems, locale specific manpaths, and combinations of both,
    if
51 # they exist, otherwise the permissions of the user running man/mandb wi
    ll
52 # be used to manipulate the manual pages. Also, mandb will not initialis
    e
53 # the database cache for any manpaths not mentioned below unless explici
    tly
54 # requested to do so.
55 #
56 # In a per-user configuration file, this directive only controls the
57 # location of catpaths and the creation of database caches; it has no ef
    fect
replace with MAN (y/n/a/q/l/^E/^Y)?

```

图 6 使用：50,100s/man/MAN/gc 命令
(7) 修改完之后，再全部恢复。

图 7

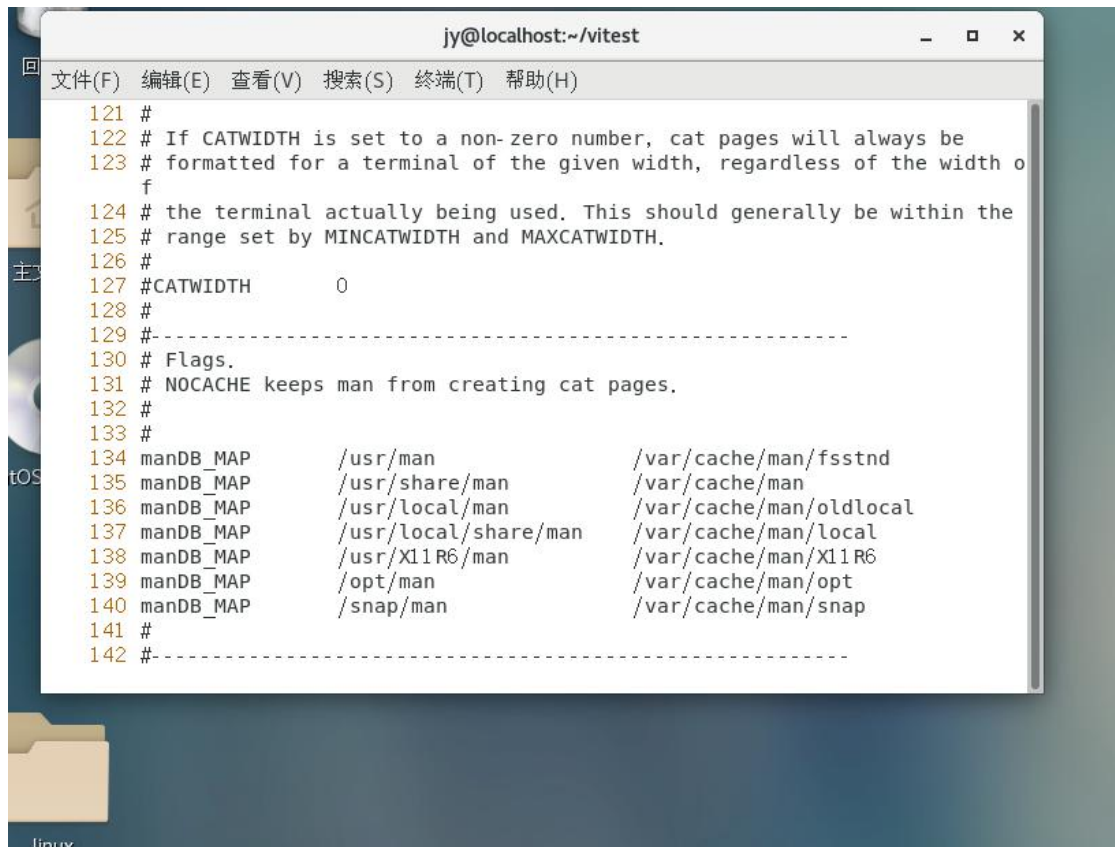
```

jy@localhost:~/vitest
文件(F) 编辑(E) 查看(V) 搜索(S) 终端(T) 帮助(H)
43 # For a manpath element to be treated as a system manpath (as most of th
    ose
44 # above should normally be), it must be mentioned below. Each line may h
    ave
45 # an optional extra string indicating the catpath associated with the
46 # manpath. If no catpath string is used, the catpath will default to the
47 # given manpath.
48 #
49 # You *must* provide all system manpaths, including manpaths for alterna
    te
50 # operating systems, locale specific MANpaths, and combinations of both,
    if
51 # they exist, otherwise the permissions of the user running MAN/MANdb wi
    ll
52 # be used to MANipulate the MANual pages. Also, MANdb will not initialis
    e
53 # the database cache for any MANpaths not mentioned below unless explici
    tly
54 # requested to do so.
55 #
56 # In a per-user configuration file, this directive only controls the
57 # location of catpaths and the creation of database caches; it has no ef
    fect
replace with man (y/n/a/q/l/^E/^Y)?

```

图 7 使用：在末行模式下输入：50,100s/MAN/man/gc
(8) 复制第 66~75 行这 10 行的内容，到最后一行之后。

图 8

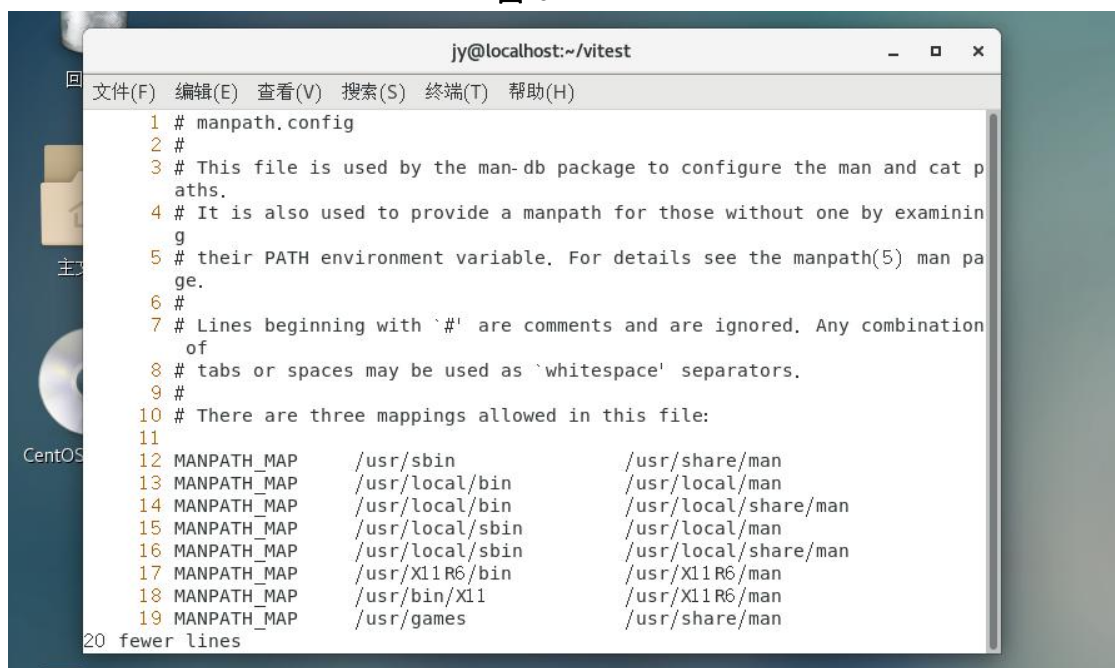


```
121 #
122 # If CATWIDTH is set to a non-zero number, cat pages will always be
123 # formatted for a terminal of the given width, regardless of the width o
124 # the terminal actually being used. This should generally be within the
125 # range set by MINCATWIDTH and MAXCATWIDTH.
126 #
127 #CATWIDTH      0
128 #
129 #-----
130 # Flags.
131 # NOCACHE keeps man from creating cat pages.
132 #
133 #
134 manDB_MAP      /usr/man                /var/cache/man/fsstnd
135 manDB_MAP      /usr/share/man          /var/cache/man
136 manDB_MAP      /usr/local/man          /var/cache/man/oldlocal
137 manDB_MAP      /usr/local/share/man    /var/cache/man/local
138 manDB_MAP      /usr/X11R6/man          /var/cache/man/X11R6
139 manDB_MAP      /opt/man                /var/cache/man/opt
140 manDB_MAP      /snap/man               /var/cache/man/snap
141 #
142 #-----
```

图 8 在命令模式下先按 65G 跳到 65 行，再按 10yy 复制 10 行的字符，按 G 到最后一行，再按 P 键粘贴

(9)删除第 11~30 行。

图 9



```
1 # manpath.config
2 #
3 # This file is used by the man-db package to configure the man and cat p
4 # It is also used to provide a manpath for those without one by examin
5 # their PATH environment variable. For details see the manpath(5) man pa
6 #
7 # Lines beginning with '#' are comments and are ignored. Any combinati
8 # tabs or spaces may be used as 'whitespace' separators.
9 #
10 # There are three mappings allowed in this file:
11 MANPATH_MAP    /usr/sbin                /usr/share/man
12 MANPATH_MAP    /usr/local/bin           /usr/local/man
13 MANPATH_MAP    /usr/local/bin           /usr/local/share/man
14 MANPATH_MAP    /usr/local/sbin          /usr/local/man
15 MANPATH_MAP    /usr/local/sbin          /usr/local/share/man
16 MANPATH_MAP    /usr/X11R6/bin           /usr/X11R6/man
17 MANPATH_MAP    /usr/bin/X11             /usr/X11R6/man
18 MANPATH_MAP    /usr/games               /usr/share/man
19
20 fewer lines
```

图 9 在命令模式下按 11G 到 11 行，按 20dd 删除了 11-30 行

(10)将这个文件另存成一个名为 manpath. test. config 的文件。

图 10

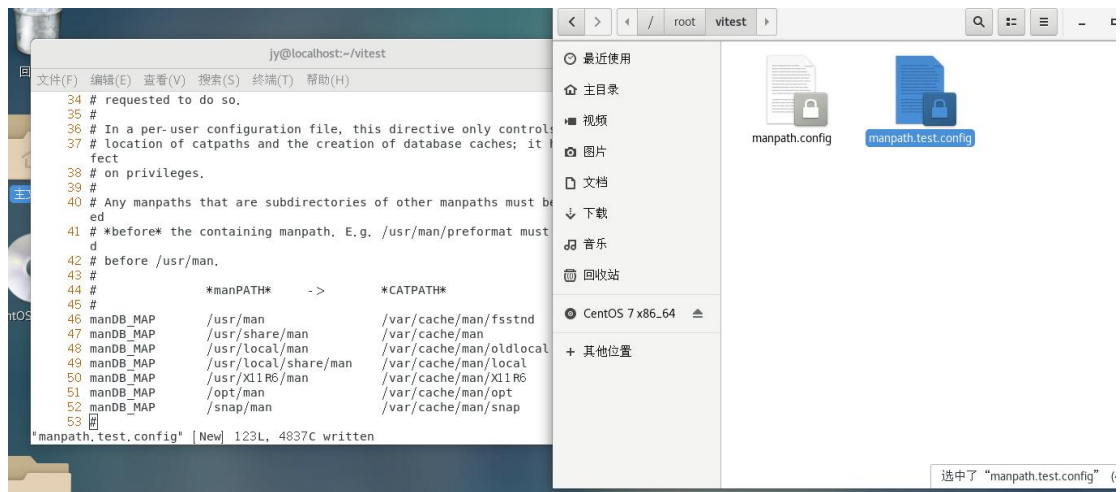


图 10 末行模式下输入: w 文件名, 另存

(11)将光标移到第 29 行, 并且删除第 15 个字符。

图 11

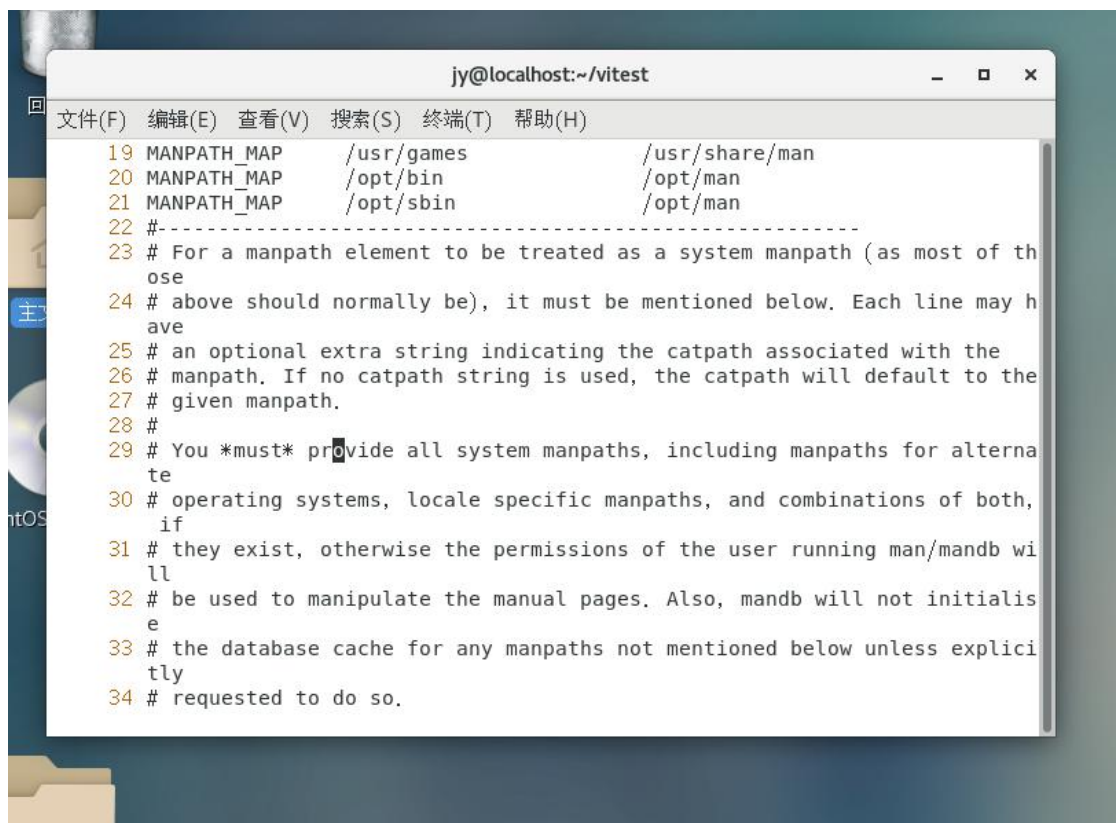
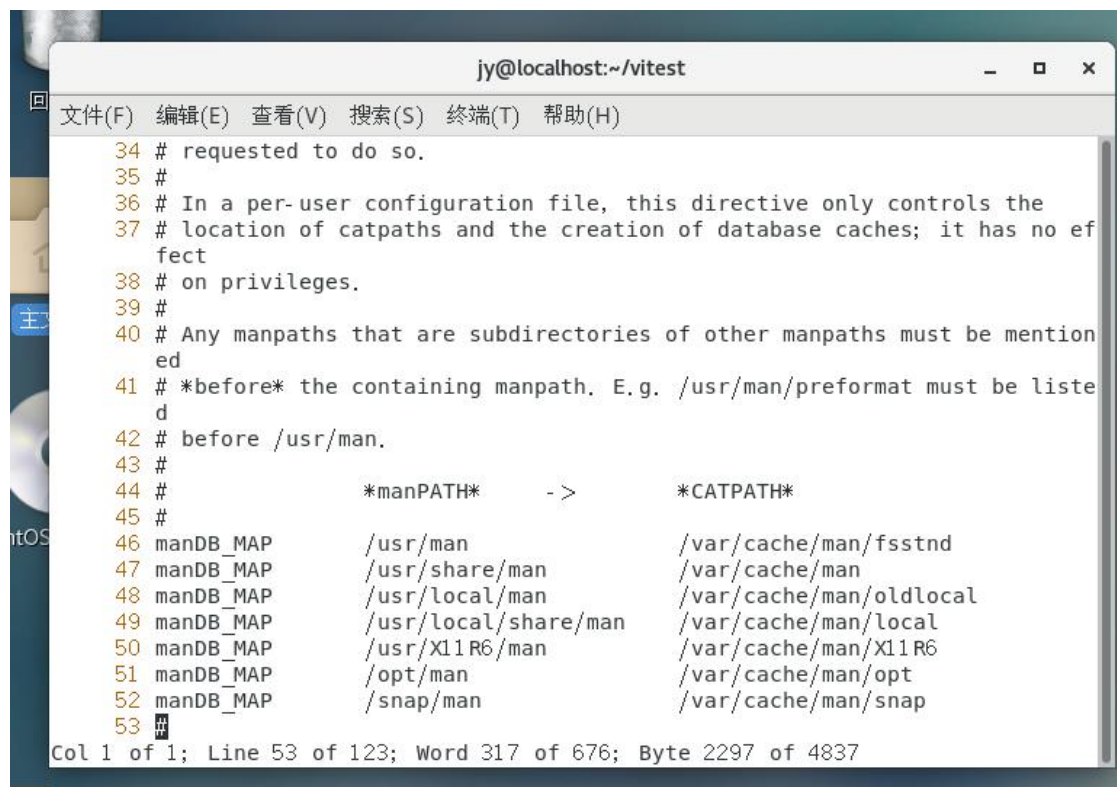


图 11 按 29G 到 29 行, 再按 15l 到第 15 个字符, 再按 i 进入插入模式删除字符

(12)统计目前的文件有多少行以及多少字符。

图 12

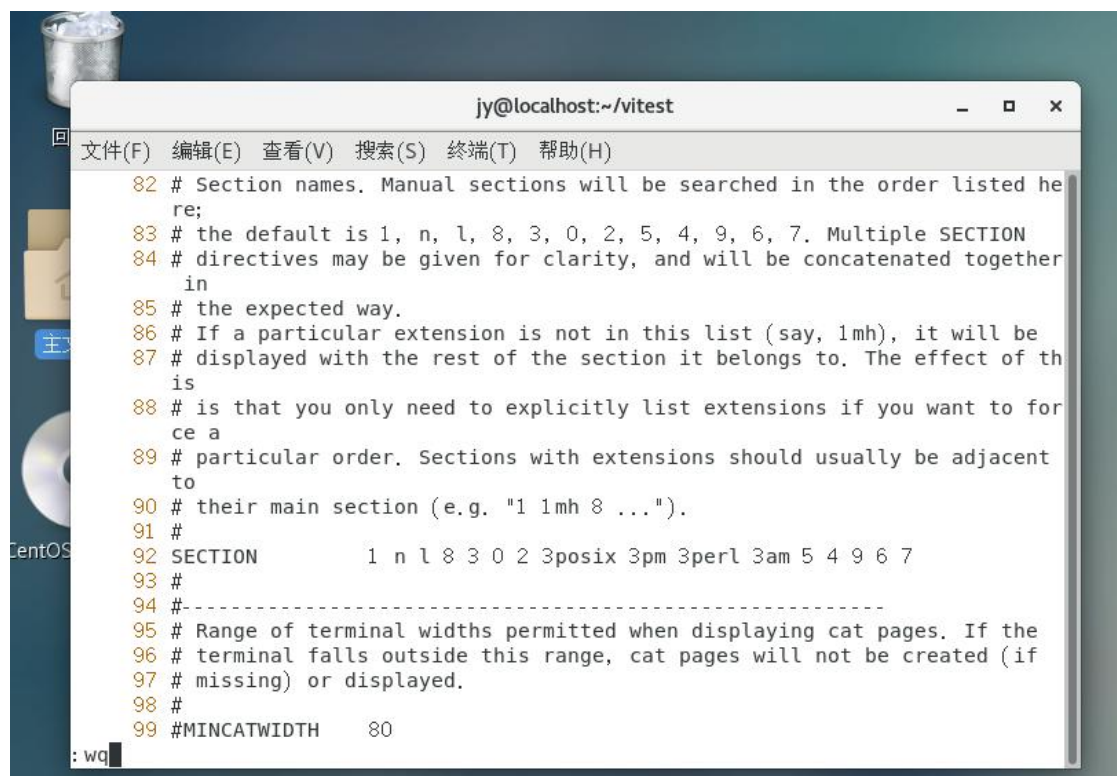


```
34 # requested to do so.
35 #
36 # In a per-user configuration file, this directive only controls the
37 # location of catpaths and the creation of database caches; it has no ef
fect
38 # on privileges.
39 #
40 # Any manpaths that are subdirectories of other manpaths must be mention
ed
41 # *before* the containing manpath. E.g. /usr/man/preformat must be liste
d
42 # before /usr/man.
43 #
44 #          *manPATH*      ->      *CATPATH*
45 #
46 manDB_MAP      /usr/man          /var/cache/man/fsstnd
47 manDB_MAP      /usr/share/man     /var/cache/man
48 manDB_MAP      /usr/local/man     /var/cache/man/oldlocal
49 manDB_MAP      /usr/local/share/man /var/cache/man/local
50 manDB_MAP      /usr/X11R6/man     /var/cache/man/X11R6
51 manDB_MAP      /opt/man           /var/cache/man/opt
52 manDB_MAP      /snap/man          /var/cache/man/snap
53 #
```

Col 1 of 1; Line 53 of 123; Word 317 of 676; Byte 2297 of 4837

图 12 再 G 到第一行，在按 ctrl+g 统计
(13)保存退出。

图 13



```
82 # Section names. Manual sections will be searched in the order listed he
re;
83 # the default is 1, n, l, 8, 3, 0, 2, 5, 4, 9, 6, 7. Multiple SECTION
84 # directives may be given for clarity, and will be concatenated together
in
85 # the expected way.
86 # If a particular extension is not in this list (say, 1mh), it will be
87 # displayed with the rest of the section it belongs to. The effect of th
is
88 # is that you only need to explicitly list extensions if you want to for
ce a
89 # particular order. Sections with extensions should usually be adjacent
to
90 # their main section (e.g. "1 1mh 8 ...").
91 #
92 SECTION      1 n l 8 3 0 2 3posix 3pm 3perl 3am 5 4 9 6 7
93 #
94 #-----
95 # Range of terminal widths permitted when displaying cat pages. If the
96 # terminal falls outside this range, cat pages will not be created (if
97 # missing) or displayed.
98 #
99 #MINCATWIDTH  80
```

: wq

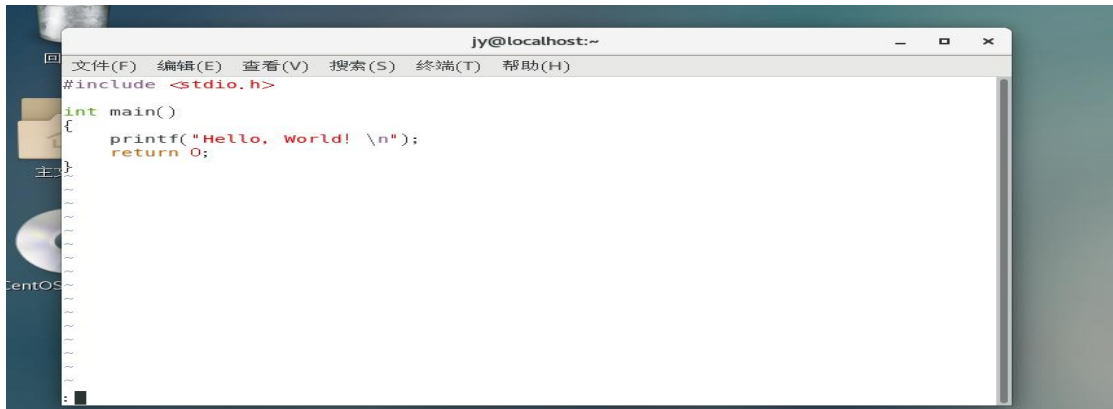
图 13 在末行模式输入 wq 保存退出

2. 实验内容

(1)在 vi 中使用 C 语言编写一个 hello world 程序,用 gcc 编译

它并运行。

图 14

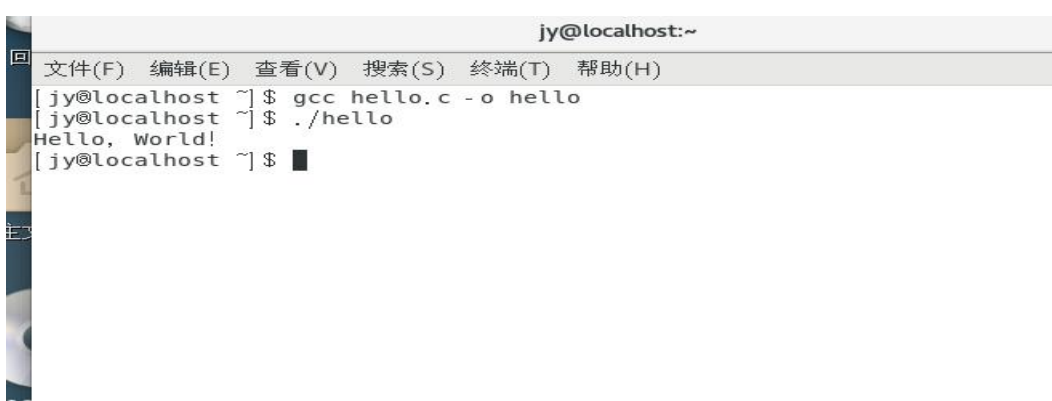
A screenshot of a vim editor window. The title bar shows 'jy@localhost:~'. The menu bar includes '文件(F)', '编辑(E)', '查看(V)', '搜索(S)', '终端(T)', and '帮助(H)'. The code in the editor is:

```
#include <stdio.h>

int main()
{
    printf("Hello, World! \n");
    return 0;
}
```

图 14 vim 编写了一个 C 语言程序

图 15

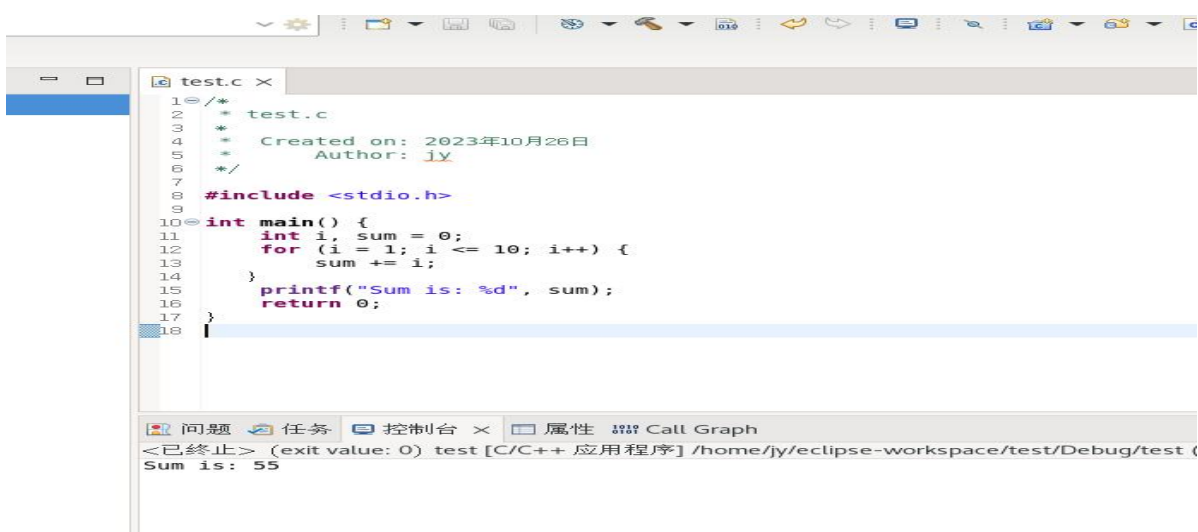
A screenshot of a terminal window. The title bar shows 'jy@localhost:~'. The menu bar includes '文件(F)', '编辑(E)', '查看(V)', '搜索(S)', '终端(T)', and '帮助(H)'. The terminal shows the following commands and output:

```
[jy@localhost ~]$ gcc hello.c -o hello
[jy@localhost ~]$ ./hello
Hello, World!
[jy@localhost ~]$
```

图 15 将 hello.c 这个 C 语言源代码文件编译成可执行文件，并命名为 hello，再在当前目录中查找名为 hello 的可执行文件并运行它

(2) 在 Eclipse 中使用 C 语言编写一个循环程序,用 Eclipse 编译并运行。使用 Eclipse 的调试功能,监视循环变量的变化情况。

图 16

A screenshot of the Eclipse IDE. The title bar shows 'test.c x'. The menu bar includes '问题', '任务', '控制台 x', '属性', and 'Call Graph'. The code in the editor is:

```
1 /*
2  * test.c
3  *
4  * Created on: 2023年10月26日
5  * Author: jy
6  */
7
8 #include <stdio.h>
9
10 int main() {
11     int i, sum = 0;
12     for (i = 1; i <= 10; i++) {
13         sum += i;
14     }
15     printf("Sum is: %d", sum);
16     return 0;
17 }
18
```

The console output at the bottom shows:

```
<已终止> (exit value: 0) test [C/C++ 应用程序] /home/jy/eclipse-workspace/test/Debug/test
Sum is: 55
```

图 16 在 eclipse 创建了一个 C 语言项目，然后运行

图 17

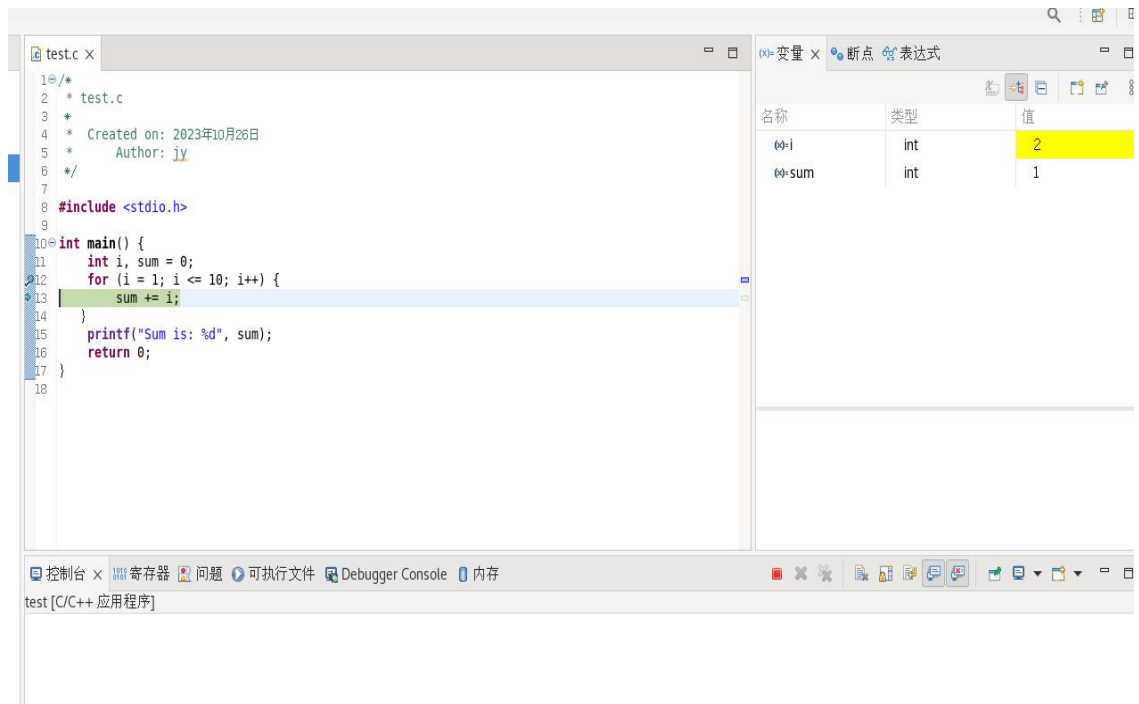


图 17 调试了这个代码，然后调试，观察 for 循环每次循环的数字变化